

S3 First Term Exam 2025
Mathematics (Paper 2)

Time Allowed: 1 hour

Instructions:

(I) There are 30 questions in this paper and each question carries equal mark. Answer **ALL** questions.

(II) The diagrams in this paper may not be drawn to scale.

1. Factorize $2x^2 + xy - 6y^2$.

A. $(2x - 3y)(x + 2y)$

B. $(2x + 3y)(x - 2y)$

C. $2(x - 3y)(x + y)$

D. $2(x + 3y)(x - y)$

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2. Factorize $2p^2 + 3pq - 2q^2 + 3p + 6q$.

A. $(p + 2q)(2p + q - 3)$

B. $(p + 2q)(2p - q + 3)$

C. $(p - 2q)(2p + q - 3)$

D. $(p - 2q)(2p - q + 3)$

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3. Which of the following have $x - 3$ as a factor?

I. $x^2 - 2x - 3$

II. $2x^2 - 11x + 15$

III. $3x^2 - 18x + 27$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

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4. If $4x^2 + px - 15$ can be factorized as $(4x + a)(x + b)$, where a and b are integers, which of the following are the possible values of p ?

I. 4

II. 17

III. 59

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

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5. $\frac{1}{x-4} - \frac{x}{x^2-3x-4} =$

A. $\frac{1}{(x+1)(x-4)}$

B. $\frac{x}{(x+1)(x-4)}$

C. $\frac{x-1}{(x+1)(x-4)}$

D. $\frac{2x-1}{(x+1)(x-4)}$

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6. $\frac{(2a^{-2})^4}{(4a)^{-3}} =$

A. $\frac{128}{a^5}$

B. $\frac{1024}{a^5}$

C. $\frac{128}{a^{11}}$

D. $\frac{1024}{a^{11}}$

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7. $12^{3n} \times 8^n =$

A. 24^n

B. 24^{3n}

C. 1728^n

D. 1728^{3n}

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8. If $a > 1$, which of the following must be true?

A. $(a^{-1})^0 = 0$

B. $a^{-n} = -a^n$

C. $a^3 \times a^5 = a^{15}$

D. $(2a)^5 = 32a^5$

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9. $27^{100} \left(-\frac{1}{3} \right)^{299} =$

A. -3

B. $-\frac{1}{3}$

C. $\frac{1}{3}$

D. 3

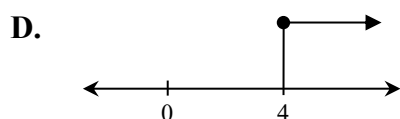
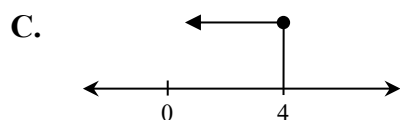
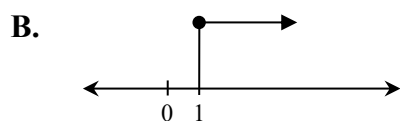
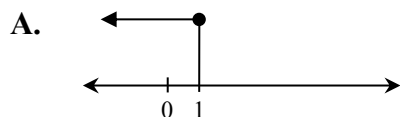
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NF 10. Convert the denary number $2^8 + 2^6 + 15$ to a binary number.

- A. 10100111_2
- B. 10101111_2
- C. 101001111_2
- D. 101011111_2



11. Which of the following diagrams represents the solutions of the inequality $3 - 2x \leq -5$?



12. If $x < -\frac{15}{4}$, which of the following CANNOT be a value of x ?

- A. -3
- B. -4
- C. $-\frac{17}{4}$
- D. $-\frac{19}{4}$



13. If a and b are non-zero numbers with $a > b$, which of the following must be true?

- A. $a - b < 0$
- B. $-a > -b$
- C. $(-2)^2 a > (-2)^2 b$
- D. $\frac{1}{a} < \frac{1}{b}$



14. If x is an integer that satisfies the inequality $\frac{x}{5} - \frac{x}{4} > 2$, then the greatest possible value of x is
- A. -41 .
B. -40 .
C. 39 .
D. 40 .
- ☐

15. If $a < b$ and $k < 0$, which of the following must be true?

I. $a^2 < b^2$

II. $k - a > k - b$

III. $\frac{a}{k^2} < \frac{b}{k^2}$

- A. I and II only
B. I and III only
C. II and III only
D. I, II and III
- ☐

16. The value of a smartphone depreciates by 40% per year. If the value of the smartphone is \$2880 now, find the value of the smartphone 2 years ago.

- A. \$460.8
B. \$1036.8
C. \$8000
D. \$18 000
- ☐

17. If the base and the height of a triangle are decreased by 12% and increased by 10% respectively, the percentage change in its area is

- A. -3.2% .
B. -2% .
C. $+0.8\%$.
D. $+2\%$.
- ☐

18. A sum of \$80 000 is deposited at an interest rate of 6% p.a. for 4 years, compounded half-yearly. Find the interest correct to the nearest dollar.

- A. \$19 200
B. \$20 998
C. \$21 342
D. \$21 639
- ☐

19. In a company, 40% of the employees are female and $x\%$ of the female employees are married. It is given that 50% of the male employees are married. If the number of employees who are married and the number of female employees are equal, find x .

A. 20
B. 25
C. 30
D. 35



20. Refer to the following progressive tax rates.

Net chargeable income	Tax rate
On the first \$50 000	2%
On the next \$50 000	6%
On the next \$50 000	10%
On the next \$50 000	14%
Remainder	17%

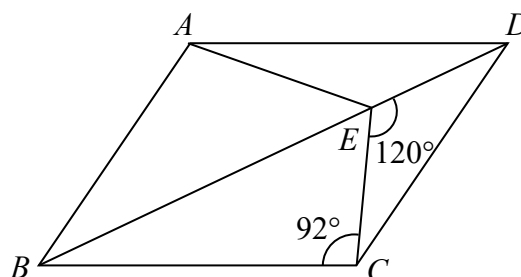
Tiffany's monthly income is \$21 000. If her allowance is \$74 000, how much salaries tax should she pay?

A. \$10 680
B. \$12 920
C. \$17 800
D. \$24 920



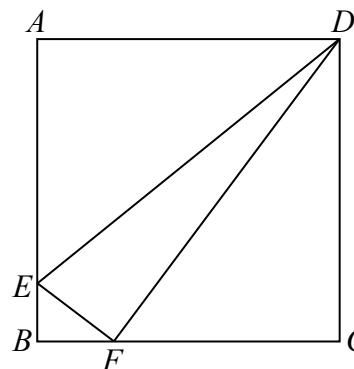
21. In the figure, BED is a diagonal of the parallelogram $ABCD$. If $\angle ECB = 92^\circ$ and $\angle CED = 120^\circ$, find $\angle ADB$.

A. 28°
B. 32°
C. 44°
D. 58°



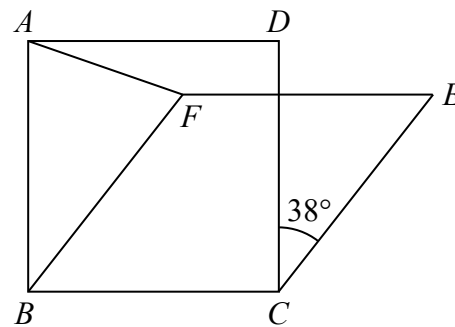
22. In the figure, $ABCD$ is a square. Let E and F be points lying on AB and BC respectively such that $FC = 3BF$ and $\angle EFD = 90^\circ$. If the area of $ABCD$ is 6400 cm^2 , then the area of $\triangle DEF$ is

A. 1200 cm^2 .
B. 1250 cm^2 .
C. 1400 cm^2 .
D. 1500 cm^2 .



23. In the figure, $ABCD$ is a square and $BCEF$ is a rhombus. If $\angle DCE = 38^\circ$, find $\angle AFB$.

- A. 71°
 B. 74°
 C. 76°
 D. 80°



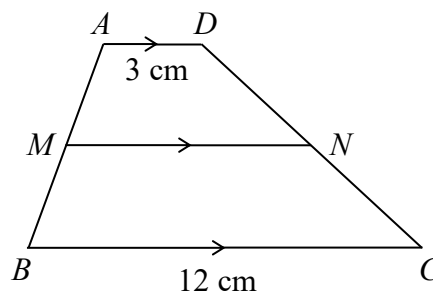
24. Which of the following description of quadrilaterals may not be correct?

- A. The diagonals of a rhombus are perpendicular to each other.
 B. The angle between a diagonal and a side of a square is 45° .
 C. The diagonals of a rectangle bisect each other into four equal parts.
 D. The interior angles of a parallelogram are bisected by the diagonals.



- NF** 25. In the figure, AMB and DNC are straight lines, $AD \parallel MN \parallel BC$ and M is the mid-point of AB . If $AD = 3$ cm and $BC = 12$ cm, find the length of MN .

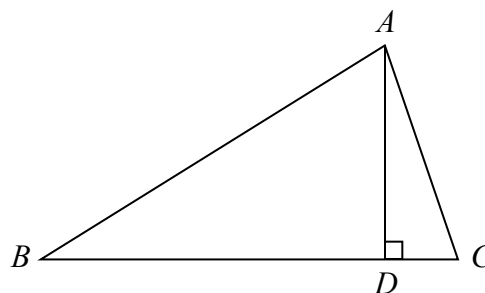
- A. 4.5 cm
 B. 6 cm
 C. 7.5 cm
 D. 9 cm



26. In the figure, BDC is a straight line and $AD \perp BC$.

Which of the following must be true?

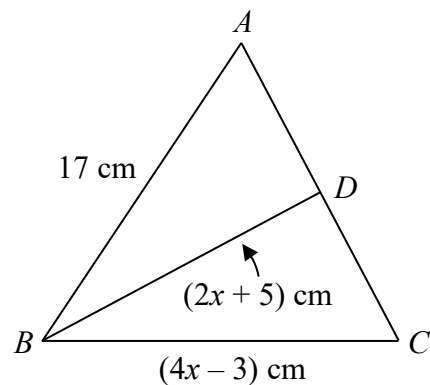
- A. AD is an angle bisector of $\triangle ABC$.
 B. AD is a perpendicular bisector of $\triangle ABC$.
 C. AD is a median of $\triangle ABC$.
 D. AD is an altitude of $\triangle ABC$.



27. In the figure, D is the mid-point of AC and $BD \perp AC$.

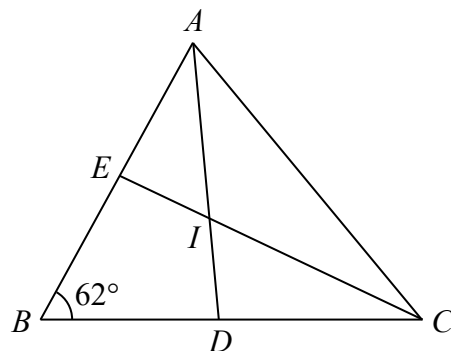
It is given that $AB = 17$ cm, $BC = (4x - 3)$ cm and $BD = (2x + 5)$ cm. Find the length of AC .

- A. 14 cm
- B. 15 cm
- C. 16 cm
- D. 17 cm



- NF** 28. In the figure, I is the incentre of $\triangle ABC$. AI and CI are produced to meet BC and AB at D and E respectively. If $\angle ABC = 62^\circ$, find $\angle EID$.

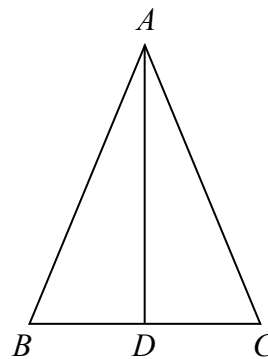
- A. 118°
- B. 121°
- C. 130°
- D. 146°



- NF** 29. In the figure, $\triangle ABC$ is an acute-angled triangle. AD is a perpendicular bisector of $\triangle ABC$. Which of the following points must lie on AD ?

- I. The incentre of $\triangle ABC$
- II. The centroid of $\triangle ABC$
- III. The orthocentre of $\triangle ABC$

- A. I and II only
- B. II and III only
- C. I and III only
- D. I, II and III



30. If $\triangle ABC$ is a right-angled triangle with $\angle ABC = 90^\circ$, which of the following are true?

- I. The circumcentre of $\triangle ABC$ lies on AC .
- II. The centroid of $\triangle ABC$ lies inside $\triangle ABC$.
- III. The orthocentre of $\triangle ABC$ lies on B .

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III



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